

Emergency Response Workshop: Saving Lives Through Smart Decisions

A 2-Hour Experiential Learning Journey

Workshop Overview

Duration: 2 hours (including 15-minute break) **Participants:** High school students (groups of 5) **Theme:** Ambulance dispatch and location optimization

Learning Objectives

Understand how real-world problems translate into decision models

Visualize feasible solutions through interactive activities

Experience the impact of constraints and randomness on outcomes

Discover concepts of optimal vs. good-enough solutions

Schedule

0:00-0:15 - Introduction & Case Presentation **0:15-1:00** - Station Rotations (5 stations × 9 minutes) **1:00-1:15** - BREAK **1:15-1:45** - Final Challenge & Group Work **1:45-2:00** - Debrief & Reflection

Opening Presentation (15 minutes)

"The City Emergency Challenge"

Setting the Scene: "Welcome to Metro City! Population: 500,000. Your mission: Design an emergency response system that saves the most lives possible. But here's the catch - you have limited resources and every second counts."

The Problem:

Metro City has 5 districts with different population densities

You have only 10 ambulances to position across the city

Emergency calls come in randomly throughout the day

Response time goal: reach any emergency within 8 minutes

Budget constraints limit where you can build stations

What OR Professionals Do:

Transform messy real-world problems into clear decisions

Balance competing objectives (cost vs. coverage)

Account for uncertainty (where will emergencies happen?)

Find the "best" solution among millions of possibilities

Station Activities (45 minutes total)

Station 1: "Map the City" - Understanding the Problem Space

Time: 10 minutes (9 min activity + 1 min transition)

Materials:

Large city map with grid overlay

Population density cards for each district

Colored pins for potential station locations

Timer showing "response radius" circles

Activity: Students place ambulance stations on the map trying to cover most of the population. They discover that some locations cover more people but cost more (downtown vs. suburbs).

Game Element: Roll dice to determine which districts have "rush hour" (doubles response time). Students must adapt their placement strategy.

Learning Focus:

Visualizing feasible locations

Understanding coverage vs. cost trade-offs

Station 2: "Racing Against Time" - Response Time Simulation

Time: 10 minutes

Materials:

Stopwatch

Emergency call cards with locations

Toy ambulances

Road network board with traffic lights

Activity: Students dispatch ambulances to emergency calls, physically moving pieces while timer runs. They experience how distance and traffic affect response times.

Game Element: Draw "traffic event" cards that create delays. Students must reroute or dispatch different ambulances.

Learning Focus:

Time as a constraint

Impact of uncertainty on decisions

Station 3: "Budget Balancer" - Resource Allocation

Time: 10 minutes

Materials:

Budget tokens (\$1M each)

Station cost cards (Downtown: \$5M, Suburb: \$3M, Rural: \$2M)

Ambulance tokens (cost \$500K each)

Performance scoreboard

Activity: With limited budget, students must decide: More stations with fewer ambulances each? Or fewer stations with more ambulances?

Game Element: Spin wheel for "city council decisions" that can increase/decrease budget mid-game.

Learning Focus:

Resource constraints shape feasible solutions

Multiple objectives (coverage, response time, cost)

Station 4: "Peak Hour Panic" - Demand Patterns

Time: 10 minutes

Materials:

Hour-by-hour demand charts

Shift scheduling boards

Ambulance availability trackers

Emergency frequency dice

Activity: Students schedule ambulance shifts to match expected demand patterns, then test their schedules against random emergency patterns.

Game Element: Roll multiple dice to generate emergency calls each "hour." Some hours overwhelm the system.

Learning Focus:

Matching resources to variable demand

Understanding peak vs. off-peak optimization

Station 5: "What If?" - Scenario Planning

Time: 10 minutes

Materials:

Scenario cards (natural disaster, major accident, sports event)

Flexible planning board

Backup resource tokens

Impact measurement cards

Activity: Students adapt their "optimal" plan to handle special scenarios, discovering how robust their solutions are.

Game Element: Draw random "crisis cards" that drastically change the problem. Teams compete to adapt fastest.

Learning Focus:

Local vs. global optimality

Robustness of solutions

Break (15 minutes)

Final Challenge (30 minutes)

"The Ultimate Test"

Setup: All groups receive the same complex scenario combining elements from all stations.

Scenario: "It's New Year's Eve in Metro City. You must plan ambulance deployment for the busiest night of the year with:

Expected 3x normal emergency calls

Major events in 3 districts

Road closures for celebrations

\$2M emergency budget boost"

Process:

Groups have 20 minutes to develop their plan

Each group presents their solution approach (2 min each)

Instructor runs simulation of each plan

Interactive Element: Use digital slider bars (or physical sliders) to show:

How changing station locations affects coverage area

How adding ambulances changes response capability

How budget constraints limit feasible solutions

Debrief & Reflection (15 minutes)

Group Discussion Questions

Understanding Translation to Models:

"What assumptions did you make about the city?"

"How did you simplify the real problem?"

Comprehending Feasible Space:

"Draw your feasible solutions - where could you place stations?"

"What made some solutions impossible?"

Understanding Optimality:

"Was there a 'perfect' solution? Why or why not?"

"How did your 'best' solution change with different scenarios?"

Key Takeaways Board

Students write one insight on sticky notes:

Assumptions: "We assumed all ambulances travel at same speed"

Constraints: "Budget limited our station choices"

Trade-offs: "Better coverage meant higher costs"

Uncertainty: "Random events changed our best plan"

Materials Checklist

Per Station:

Instruction cards with clear visuals

Game materials (dice, cards, tokens)

Recording sheets for results

Timer/stopwatch

General Supplies:

Large Metro City maps (1 per group)

Whiteboards for calculations

Sticky notes and markers

Digital/physical sliders for final activity

Prize tokens for gamification

Facilitator Notes

Volunteer Roles:

Station Leaders (5): Guide activities, ensure understanding

Floater: Helps struggling groups, manages time

Tech Support: Manages final presentation/simulation

Key Messages to Reinforce:

There's rarely one "right" answer in real decisions

Models help us think systematically about complex problems

Small changes in assumptions can lead to very different solutions

Randomness and uncertainty are part of real life

Assessment Opportunities:

Observe group problem-solving approaches

Listen to mathematical thinking without formal notation

Note how students handle trade-offs

Track evolution of understanding through stations

Learning Objectives

Students should leave being able to:

- ✓ Explain why some ambulance locations are better than others
- ✓ Describe how constraints limit choices
- ✓ Understand why "optimal" depends on what you're optimizing for
- ✓ Recognize that randomness affects real-world decisions
- ✓ See mathematics as a tool for making better decisions